

Mutual Fund Analytics Platform

Bluestock Fintech Capstone Project

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Prepared by: M Abhinaya Reddy

Company: Bluestock Fintech Pvt. Ltd.

Domain: Mutual Fund / Fintech

Duration: 7 Working Days | ~50-55 Hours

Date: June 2026

GitHub: github.com/abhinay1-11/bluestock_mf_capstone

1. Executive Summary

This capstone project builds a full-stack Mutual Fund Analytics Platform for Bluestock Fintech Pvt. Ltd. The platform ingests publicly available data from AMFI India and mfapi.in, transforms it through a robust ETL pipeline, stores it in a relational SQLite database, and presents actionable insights via an interactive Power BI dashboard.

The Indian mutual fund industry has grown at a remarkable pace. As of December 2025, the industry manages over Rs. 81 lakh crore in AUM across 26.12 crore investor folios. Monthly SIP inflows crossed an all-time high of Rs. 31,002 crore in December 2025. This platform provides data-driven tools to navigate this growing landscape.

Key Outcomes

- Loaded and cleaned 10 datasets covering 87,000+ rows of real MF data
- Built a 5-table SQLite star schema database with full referential integrity
- Produced 9 publication-quality EDA charts across NAV, AUM, SIP, and demographics
- Computed Sharpe ratio, Sortino, Alpha, Beta, CAGR, VaR, and Max Drawdown for 40 funds
- Delivered a 4-page interactive Power BI dashboard with slicers and drill-through
- Built a fund recommender system based on investor risk appetite
- Identified that 97.8% of SIP investors are at risk of discontinuation

2. Problem Statement

Despite the massive growth of India's mutual fund industry, individual investors and financial advisors often struggle to make data-driven fund selection decisions due to fragmented data, lack of unified analytics platforms, and complex financial metrics.

P1: Data Fragmentation

NAV data, AUM data, SIP flow data and portfolio holdings are available on different sections of the AMFI website in different formats. No single unified database exists. This project consolidates all data into a single SQLite database via an automated ETL pipeline.

P2: Performance Comparison Gap

Investors cannot easily compare multiple funds across different AMCs on a risk-adjusted basis. Raw NAV data requires significant transformation to compute Sharpe ratio, Alpha, Beta, etc. This project computes and visualises all key risk-return metrics in a single dashboard.

P3: No Benchmark Tracking

Most retail investors do not know whether their fund is outperforming its benchmark index. This project joins NAV data with Nifty 50 and Nifty 100 benchmark prices and computes rolling alpha and tracking error for the top 5 funds.

P4: Investor Behaviour Blind Spot

AMCs and distributors have limited visibility into how demographic and geographic factors influence SIP amounts and redemption patterns. This project analyses investor transaction data to generate demographic segmentation and behavioural insights.

3. Data Sources & Datasets

All datasets used in this project are derived from publicly available, freely accessible sources including AMFI India (amfiindia.com), mfapi.in REST API, and NSE/BSE public data.

Dataset	Rows	Description
01_fund_master.csv	40	Master list of 40 mutual fund schemes with AMFI codes, fund house, category, expense ratio, risk grade
02_nav_history.csv	46,000	Daily NAV for all 40 schemes from Jan 2022 to May 2026, anchored to real mfapi.in values
03_aum_by_fund_house.csv	90	Quarterly AUM in Rs. crore for 10 fund houses from 2022 to 2025
04_monthly_sip_inflows.csv	48	Month-wise SIP inflow, active SIP accounts, new registrations and SIP AUM
05_category_inflows.csv	144	Net inflows by fund category for FY 2024-25
06_industry_folio_count.csv	21	Total MF folios broken by Equity, Debt, Hybrid from 2022 to 2025
07_scheme_performance.csv	40	1yr/3yr/5yr returns, Sharpe, Sortino, Alpha, Beta, Max Drawdown
08_investor_transactions.csv	32,778	SIP, Lumpsum and Redemption transactions for 5,000 investors
09_portfolio_holdings.csv	322	Top equity holdings for each fund as of Dec 2025
10_benchmark_indices.csv	8,050	Daily closing values for Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap

4. System Architecture & ETL Pipeline

The project follows a classic data engineering architecture: Extract -> Transform -> Load -> Analyse -> Visualise. This mirrors real-world fintech data pipelines used at companies like Zerodha, Groww, and Paytm Money.

Layer 1 - Extract

- 10 CSV datasets provided by Bluestock Fintech (AMFI-sourced)
- Live NAV data fetched from mfapi.in REST API for 6 key schemes
- API endpoint: GET https://api.mfapi.in/mf/{scheme_code}

Layer 2 - Transform

- Parsed date strings to proper datetime objects
- Forward-filled missing NAV values for weekends and holidays using `resample('D').ffill()`
- Validated AMFI codes - confirmed 100% coverage between `fund_master` and `nav_history`
- Standardised transaction types to SIP / Lumpsum / Redemption
- Validated NAV > 0, `amount_inr` > 0, expense ratio within 0.1-2.5% range

Layer 3 - Load

- Designed 5-table SQLite star schema: `dim_fund`, `dim_date`, `fact_nav`, `fact_transactions`, `fact_performance`
- Loaded all cleaned data using SQLAlchemy `create_engine` + `df.to_sql()`
- Total records loaded: 40 funds, 64,320 NAV records, 32,778 transactions

Layer 4 - Analyse

- EDA notebooks: 9 charts across NAV trends, AUM growth, SIP flows, demographics
- Performance metrics: CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR
- Advanced analytics: Rolling Sharpe, cohort analysis, SIP continuity, sector HHI

Layer 5 - Visualise

- 4-page Power BI dashboard with slicers, KPI cards, and drill-through
- Pages: Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends

5. EDA Findings

5.1 NAV Trends

All 40 schemes showed consistent NAV growth from 2022 to 2026. A clear bull run was observed in 2023 with strong upward momentum across equity funds. Mid-2024 saw a correction phase, particularly visible in mid-cap and small-cap funds. By 2026, most equity funds had recovered and reached new highs.

5.2 AUM Growth

SBI Mutual Fund dominates at Rs. 12.5 lakh crore AUM in 2025, making it the largest AMC in India. ICICI Prudential MF (Rs. 10.74L Cr) and HDFC Mutual Fund (Rs. 9.30L Cr) are second and third respectively. All 10 fund houses showed consistent AUM growth from 2022 to 2025.

5.3 SIP Inflow Trends

Monthly SIP inflows grew nearly 3x from Rs. 11,500 crore in January 2022 to an all-time high of Rs. 31,002 crore in December 2025. This reflects India's deepening equity investment culture driven by digitalisation and increased financial awareness among millennials.

5.4 Investor Demographics

The 26-35 age group accounts for 41.1% of all investors, confirming that millennials are the primary driver of retail MF growth. Male investors account for 66.5% of transactions vs 33.5% female, indicating significant room for growth in female investor participation. Madhya Pradesh leads all states in total SIP investment amount.

5.5 Folio Count Growth

Industry folios grew from 13.26 crore to 26.12 crore in just 4 years - a 97% increase - showing massive retail penetration across India. The T30 vs B30 split shows 66.3% from top 30 cities with a growing 33.7% contribution from beyond top 30 cities.

5.6 Sector Allocation

Banking dominates at 19.2% of all equity fund holdings, followed by IT at 13.4% and Pharma at 12%. Together these three sectors account for 44.6% of all equity fund holdings, indicating significant sector concentration risk at the industry level.

6. Fund Performance Analysis

6.1 CAGR Analysis

Axis Midcap Fund leads the 3-year CAGR rankings at 35.1%, followed by Mirae Asset Large Cap at 34.0% and HDFC Mid-Cap Opportunities at 32.4%. Liquid funds consistently deliver 6-7% CAGR, reflecting their nature as low-risk, stable instruments.

6.2 Risk-Adjusted Returns

Metric	Formula	Top Fund	Value
Sharpe Ratio	$(R_p - R_f) / \text{Std} \times \sqrt{252}$	Mirae Asset Large Cap	1.07
Sortino Ratio	$(R_p - R_f) / \text{Downside Std} \times \sqrt{252}$	Mirae Asset Large Cap	1.49
Alpha (3yr)	Intercept x 252 (OLS vs Nifty 100)	SBI Small Cap	0.30
Max Drawdown	$\min(\text{NAV} / \text{running_max} - 1)$	SBI Small Cap Direct	-52.57%
VaR 95%	5th percentile of daily returns	ABSL Small Cap	-2.39%

6.3 Fund Scorecard

The composite fund scorecard (30% x 3yr return + 25% x Sharpe + 20% x Alpha + 15% x expense ratio + 10% x max drawdown) ranks ICICI Pru Midcap Fund at #1 with a score of 85.13, followed by Axis Midcap (82.0) and HDFC Mid-Cap Opportunities (80.5). Mirae Asset Large Cap ranks 4th despite having the best Sharpe ratio, as its 3-year CAGR is slightly lower.

6.4 Benchmark Comparison

All top 5 funds by scorecard massively outperformed Nifty 50 and Nifty 100 over the 3-year period. Starting from a base of 100, the top funds grew to 220-260 by 2026 while Nifty 50 dropped below 100 during the 2024 correction. Mirae Asset Large Cap has the lowest tracking error vs Nifty 100 at 0.1897.

7. Power BI Dashboard

The interactive dashboard is built in Power BI Desktop and contains 4 report pages. All pages include slicers for filtering by fund house, category, state, and date range.

Page 1 - Industry Overview

- KPI Cards: Total AUM (Rs. 12.5L Cr), SIP Inflows (Rs. 31K Cr), Folios (26.12 Cr), Schemes (40)
- Line chart: AUM trend by fund house from 2022 to 2025
- Bar chart: AUM by AMC showing SBI dominance

Page 2 - Fund Performance

- Scatter plot: Return (X) vs Risk/StdDev (Y), bubble size = AUM
- Sortable fund scorecard table with Sharpe, Alpha, Max Drawdown
- NAV line chart vs Nifty 100 benchmark with dual Y-axis
- Slicers: Fund House, Category, Plan

Page 3 - Investor Analytics

- Horizontal bar chart: Total SIP amount by state
- Donut chart: Lumpsum 58.49%, Redemption 35.34%, SIP 6.17%
- Bar chart: Average SIP amount by age group
- Line chart: Monthly transaction volume trend

Page 4 - SIP & Market Trends

- Dual-axis: SIP inflow bars + Nifty 50 line (2022-2025)
- Matrix table: Category-wise net inflows by month
- Bar chart: Top categories by total net inflow

8. Advanced Analytics

8.1 Value at Risk (VaR)

Historical VaR at 95% confidence was computed for all 40 funds. Small Cap funds dominate the highest risk category. ABSL Small Cap Fund has the worst daily VaR of -2.39%, meaning on a bad day investors can lose up to 3% of their investment. The Conditional VaR (CVaR) shows the average loss on those worst days is even higher at -3.03%.

8.2 Investor Cohort Analysis

Investors were grouped by their first transaction year. The 2024 cohort accounts for 4,803 investors with an average SIP amount of Rs. 1.07 lakh and total invested of Rs. 349 crore. The 2025 cohort is still small at 197 investors. HDFC Mid-Cap Opportunities Fund is the most preferred fund for 2024 cohort investors.

8.3 SIP Continuity Analysis

Among investors with 6 or more SIP transactions, 97.8% (1,332 out of 1,362) are classified as at-risk with an average gap of 64.9 days between transactions - almost double the 35-day threshold. Only 30 investors maintain truly regular monthly SIPs. This is a critical finding for AMC retention strategies.

8.4 Fund Recommender

A simple rule-based fund recommender was built that takes investor risk appetite (Low / Moderate / High / Very High) and returns the top 3 funds by Sharpe ratio within the matching risk category. For Moderate risk investors, Mirae Asset Large Cap (Sharpe 1.07), SBI Bluechip (0.86), and Nippon India Large Cap (0.76) are recommended.

8.5 Sector HHI Concentration

The Herfindahl-Hirschman Index (HHI) was computed for each equity fund to measure sector concentration. Axis Bluechip Fund is the most concentrated with HHI of 2,065 (heavily weighted in IT). Nippon India Small Cap has the lowest HHI of 1,063, making it the most diversified equity fund in the dataset.

9. Limitations

- The dataset covers only 40 out of 1,908 AMFI-registered schemes - findings may not generalise to the full industry
- Investor transaction data is synthetically generated based on real demographic distributions - actual behavioural patterns may differ
- NAV values are anchored to real historical values but simulated forward - performance metrics reflect data patterns, not guaranteed future returns
- The Power BI dashboard requires Power BI Desktop for full interactivity - the exported PDF is static
- Alpha and Beta calculations show low R-squared values due to the nature of daily return noise - longer timeframes would produce more reliable regression estimates
- The fund recommender uses only Sharpe ratio as the ranking criterion - a production system would incorporate more factors including fund manager track record and AUM stability

10. Recommendations

For AMCs and Distributors

- Implement SIP reminder notifications for at-risk investors (97.8% have gaps > 35 days) - even a 10% reduction in at-risk investors could significantly improve AUM retention
- Target the 26-35 age group with digital-first SIP products as they represent 41.1% of investors
- Increase focus on female investor acquisition - at 33.5%, there is significant untapped market potential
- Expand B30 city outreach - the 33.7% B30 contribution shows growing penetration beyond metros

For Retail Investors

- Moderate risk investors should consider Mirae Asset Large Cap (Sharpe 1.07) or SBI Bluechip for stable risk-adjusted returns
- Small Cap funds should only be chosen with a 7+ year investment horizon given VaR of up to -2.39% daily
- Diversification across fund categories is recommended - fund return correlations are near zero, meaning holding multiple categories does not duplicate risk
- Banking sector concentration at 19.2% across equity funds warrants monitoring of banking sector health

For Bluestock Fintech

- Automate the ETL pipeline as a daily cron job to fetch fresh NAV data from mfapi.in every weekday at 8 PM
- Build a Streamlit web app as an alternative to Power BI for broader accessibility
- Implement Monte Carlo simulation to project NAV growth over 5 years with uncertainty bands
- Add Markowitz Efficient Frontier portfolio optimisation for fund selection

11. Deliverables Summary

#	Deliverable	Format	Status
D1	ETL Pipeline Script	data_ingestion.py, live_nav_fetch.py, run_pipeline.py	Complete
D2	SQLite Database	bluestock_mf.db + schema.sql + queries.sql	Complete
D3	EDA Notebook	EDA_Analysis.ipynb (9 charts)	Complete
D4	Performance Metrics	Performance_Analytics.ipynb + CSVs	Complete
D5	Interactive Dashboard	bluestock_mf_dashboard.pbix + Dashboard.pdf	Complete
D6	Advanced Analytics	Advanced_Analytics.ipynb + recommender.py	Complete
D7	Final Report + Slides	Final_Report.pdf + Presentation.pptx	Complete